DP-gSpan: A Pattern Growth-based Differentially Private Frequent Subgraph Mining Algorithm

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***Abstract*—Frequent subgraph mining has been one of the researches focuses in the field of data mining, which plays an important role in understanding social interaction mechanisms, urban planning, and studying the spread of diseases in social networks. Frequent subgraph mining provides a lot of valuable information. However, mining and publishing frequent subgraphs brings more and more risks of privacy leakage. In order to solve the problem of privacy leakage, the combination of frequent subgraph mining based on Apriori and differential privacy has become the mainstream method. Nevertheless, most of the existing research studies suffer from too large candidate subgraph sets, low accuracy, and low efficiency. Therefore, we propose a more secure and effective depth-first search algorithm for frequent subgraph mining under differential privacy, which is referred to as DP-gSpan. We design a heuristic truncation strategy and a new privacy budget allocation strategy to realize the reduction of the candidate set size and the rational allocation of the privacy budget. Through privacy analysis, we prove that DP-gSpan satisfies ε-differential privacy. Experimental results over a large number of real-world datasets prove that the performance of the proposed mechanism is better.**

Keywords—Frequent subgraph mining, Differential privacy, gSpan

I. INTRODUCTION

Real-world data can be easily modeled as a network graph where a set of nodes are interconnected by a relationship. Frequency subgraph mining (FSM) [1] as an important theme of data mining, determines which subgraphs have a higher value by analyzing the frequency of the appearance of subgraphs in graph transactions for various forms of decision-making in the future. It is widely used in classification, clustering, and indexing in bioinformatics, social networks, network analysis, and other fields. For example, discovering the movement trajectory of patients diagnosed with Covid-19 in the social interaction chart has a considerable effect on the study of the spread of the disease. However, if the data is sensitive (e.g., trajectory graphs, mobile phone call graphs, and web-click graphs), publishing them directly will raise the risk of privacy disclosure greatly. Therefore, what is expected now is to process the data before releasing it for social use.

In order to improve the privacy and security of data, the current privacy protection methods are based on k-anonymity [2], based on l-diversity [3], and based on t-closeness and other privacy protection technologies. These methods usually require special assumptions on the attack model. When the assumption is violated, it is difficult to provide an adequate privacy guarantee.

Dwork [4] and others proposed differential privacy(DP) technology in 2006, which ensures that an attacker with prior knowledge cannot obtain any specific information record from the published model. The idea is to convert the original data set or add an appropriate amount of noise to the statistical results to achieve the effect of privacy protection. Generally speaking, through randomization mechanisms, such as adding a carefully selected amount of disturbance noise, differential privacy ensures that the calculation output is not sensitive to changes in any single record, thereby limiting privacy leakage through the results.

We propose a novel frequent subgraph mining algorithm based on differential privacy in this paper, which is referred to as DP-gSpan. DP-gSpan consists of a heuristic truncation strategy and a new privacy budget allocation strategy to improve data utility and privacy. Different from the previous algorithms that frequent subgraph mining method based on Apriori under differential privacy, DP-gSpan is an algorithm based on pattern growth for frequent subgraph mining under differential privacy. Compared with the Apriori-based frequent subgraph mining algorithm under differential privacy in the traditional method, which generates a large number of repeated candidate subgraphs and requires multiple scans of the database and it has high time complexity. The idea of DP-gSpan under differential privacy is to implement frequent subgraph mining based on the DFS coding of the rightmost expanded graph. This method reduces the generation of a large number of redundant subgraphs during the mining process and reduces the number of isomorphism detections, so the efficiency is improved. In this approach, the heuristic truncation strategy is used to reduce the size of the candidate set to improve accuracy. The reasonable allocation of the privacy budget is realized by combining the new privacy budget allocation strategy with the characteristics of frequent subtrees.

The first challenge of the above problems is to design a heuristic truncation strategy that can effectively reduce the size of the candidate set. In this approach, pruning is performed after the candidate set is generated, and the heuristic idea is used to minimize the number of candidate sets. The second challenge is to design a privacy budget allocation strategy that can be better applied to the frequent subgraph DFS coding tree based on depth-first production. According to the characteristics of frequent subtrees, we assign different weights to each column of the DFS coding tree based on their depths, then the appropriate privacy budgets are allocated to different columns according to the different weights. The formal privacy analysis can effectively prove that DP-gSpan satisfies ε-differential privacy. The experimental results of multiple real data sets also show that DP-gSpan is significantly better than other frequent subgraph mining algorithms based on differential privacy.

The key contributions of this paper are summarized as follows:

- We present a novel two-phase algorithm for mining frequent subgraphs based on differential privacy called DP-gSpan, which can further improve the data utility while ensuring data satisfies ε-differential privacy.
- In order to better balance the privacy and data utility, this paper designs a new privacy budget allocation strategy and a heuristic truncation strategy, which further rationalizes the privacy budget allocation and reduces the size of the candidate set, respectively.
- The formal privacy analysis proves that DP-gSpan satisfies ε-differential privacy, and a large number of experiments also show that DP-gSpan has better performance.

The rest of the paper is organized as follows. Section 2 reviews related work; Section 3 presents the necessary background on differential privacy and frequent subgraph mining; Section 4 introduces the DP-gSpan algorithm proposed in this paper and its specific details; Section 5 gives the formal privacy analysis; The experimental results are reported in Section 6; Section 7 concludes the paper.

II. RELATED WORK

With the rapid development of data mining technology, the design and implementation of frequent pattern mining algorithms with privacy protection have received extensive attention. There are two settings for data analysis based on privacy protection. One is an interactive setting, which provides different privacy protection methods for different data analysis tasks, the other is a noninteractive setting, which develops privacy protection methods for synthetic data sets for data analysis. For noninteractive settings, Li et al. [6] proposed the latest technology for analyzing graphics data, which combined with anonymization technology to provide an open-source graph data sharing and publishing system, called SeGraph system. The research in this paper is mainly aimed at interactive settings.

Existing research on frequent pattern mining based on privacy protection is divided into three categories, they are frequent subgraph mining, frequent itemset mining, and frequent sequence mining. The research status of each category is as follows.

There are two types of frequent subgraph mining algorithms based on privacy protection. One is to find frequent subgraphs from a set of input graphs. The other variant of the FSM problem is to find frequent subgraphs in different regions of a single graph. For example, Shen et al. [7] proposed an algorithm based on Markov Chain Monte Carlo (MCMC) sampling to realize frequent subgraphs mining under differential privacy, but the algorithm guarantees only the weaker (ε, ξ)-differential privacy. This is differential privacy under relaxed conditions. Besides, their algorithm selects frequent subgraphs from all possible graphs directly which may or may not exist in the input graphs, and it will introduce a lot of noise. Cheng et al. [8] proposed a two-stage frequent subgraph mining algorithm. In the first phase, the binary estimation and the conditional exponential method are used to mine frequent subgraphs. The number of candidate subgraphs retained in each layer is estimated. The conditional exponential mechanism screens the candidate set and then uses the lattice-based noisy support computation approach in the second stage. This further improves the data utility and satisfies ε- differential privacy. Although the above algorithm can solve the problem of frequent subgraph mining under differential privacy, it still needs to be improved in balancing the privacy and data utility and efficiency. Different from the above-mentioned frequent subgraph mining algorithm under differential privacy, this paper proposes a depth-first frequent subgraph mining algorithm under differential privacy named DP-gSpan.

Recently, a new solution based on differential privacy was proposed to solve the problem of subgraph counting. The algorithm proposed by Karwa et al. [9] can privately calculate the set of two subgraph families (k stars and k triangles) of occurrences. Chen et al. [10] proposed a new recursive mechanism to publish a variety of subgraphs privately. A privacy protection data analysis platform wPINQ, which can be used to answer subgraph count queries is developed by Proserpio et al. [11]. Zhang et al. [12] proposed a stepped framework to calculate the number of occurrences of subgraphs under differential privacy. Ingalalli V et al. [13] proposed an approach called MuGraM that finds frequent patterns in a single large multigraph. Different from the above research, this paper is aimed at the mining process from a collection of input graphs.

Some studies combine privacy protection with other frequent pattern mining. They are frequent itemset mining and frequent sequence mining. Li et al. [14] proposed a database reconstruction-based algorithm for PPFIM (DR-PPFIM) that can not only achieve a high degree of privacy but also afford a reasonable data utility. In order to protect the privacy of datasets from frequency analysis attacks and verify the integrity of mining results, Qiu S et al. [15] proposed a series of privacy-preserving building blocks by employing Shamir's secret sharing scheme. For outsourcing databases, Li et al. [16] proposed a solution based on homomorphic encryption. When mining frequent itemsets of outsourcing data, Li et al. [17] proposed a privacy protection frequent itemset mining algorithm based on database reconstruction by using reserved insensitive frequent itemsets. Wang et al. [18] proposed a differentially private top-k closed co-occurrence patterns mining algorithm across multiple streams with exponential mechanism and Laplace mechanism. Wang et al. [19] proposed SVIM, a protocol for finding frequent items in the set-valued LDP setting. Su et al. [20] proposed a differentially private FIM algorithm based on the FP-growth algorithm, which is referred to as PFP-growth. For dense data sets, Cheng et al. [21] proposed the DP-Apriori algorithm, which limits the maximum transaction cardinality in the data set and uses the Apriori algorithm to mine all frequent itemsets. Lee et al. [22] proposed a differential privacy frequent itemset algorithm based on the FP tree. The method first constructs a noisy FP tree and then uses the FP-growth algorithm to mine frequent itemsets. Wang et al. [23] proposed the PrivSuper algorithm which directly adds all their subsets to the result of frequent itemsets. Zhang et al. [24] proposed a new personalized local privacy protection scheme for smart devices, which provides strict privacy guarantees while retaining ideal and accurate results. Xiong et al. [25] proposed a differential privacy frequent itemset mining algorithm based on large-scale data sampling and length constraint strategy, which solved the problem of privacy protection on large-scale data sets. Wang et al. [26] proposed an efficient differential privacy frequent itemset mining algorithm. Ryan et al. [27] designed a differential privacy Top-k frequent itemset mining that can face unknown data domains without any data structure algorithm.

Yan et al. [28] proposed Differential Private Frequent Sequence Mining(DPFSM) algorithm, which uses downward closure property to generate a candidate set of sequence patterns, smart truncating based technique to sample frequent patterns in the candidate set and a geometric mechanism to perturb the true supports of each sampled pattern. Xu et al. [29] discovered frequent subsequences under differential privacy, which utilizes a sampling-based candidate pruning technique.

III. PRELIMINARIES

A. Differential Privacy

Differential privacy [4] as the current mainstream privacy protection model is the only level of privacy protection currently that can be proven mathematically. It ensures that the opponent with most of the prior knowledge cannot infer the existence of a specific individual in the data set from the random output.

Definition 1: (ε-differential privacy[4]). Given two neighbor data sets D and D' (with a difference of at most one piece of data), a random algorithm A, if the algorithm A output O on D and D' satisfies the following inequality, then it satisfies ϵ-differential privacy.

$$Pr[A(D) = O] \leq exp(\varepsilon) \times Pr[A(D') = O] \quad (1)$$

Definition 2: (Sensitivity). Given a function f: $D^n \rightarrow R^d$, its sensitivity Δf is defined as follows:

$$\Delta f = max_{D,D'} \|f(D) - f(D')\|_1 \quad (2)$$

Among them, D and D' are neighbors of each other, $\| \ \|_1$ represent the first-order normal form of a vector. The sensitivity in differential privacy can be used to reflect the maximum difference in the results of the query function on a pair of adjacent data sets. The noise added by differential privacy is proportional to sensitivity.

Definition 3: (Laplace mechanism). The Laplace mechanism is a privacy protection mechanism for processing the algorithm whose output result is a numeric type. The specific concept is as follows:

For any function $f: D \rightarrow R^d$, with sensitivity Δf. If the algorithm Q satisfies the following equation, it satisfies ε-differential privacy, where $Lap(\frac{\Delta f}{\varepsilon})$ is a random noise that obeys the Laplace distribution.

$$Q(D) = f(D) + Lap(\frac{\Delta f}{\varepsilon}) \quad (3)$$

Theorem 1: (Sequential Composition). Given a database D, there is a differential privacy algorithm $M_1, M_2, \ldots, M_t$, The privacy budgets are $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_t$, Then the combined algorithm $M(M_1(D), M_2(D), \ldots, M_t(D))$ satisfies $(\sum_{i=1}^{t} \varepsilon_i)$-differential privacy.

Theorem 2: (Parallel Composition). With differential privacy algorithm $M_1, M_2, \ldots, M_t$, The privacy budgets are $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_t$, $D_1, D_2, \ldots, D_t$ are disjoint data sets, Then the combined algorithm $M\big(M_1(D_1), M_2(D_2), \ldots, M_t(D_t)\big)$ satisfies $(\max\varepsilon_i)$-differential privacy.

B. Frequent Subgraph Mining

There are two main types of frequent subgraph mining algorithms, one is the Apriori-based algorithm, the other is the pattern-growth algorithm. The main idea of the pattern-growth algorithm is to use depth-first search (DFS). Compared with the algorithm using the Apriori-based method, it reduces the generation of redundant graphs and improves the execution efficiency of the algorithm.

Graph G is defined as a tuple (V, E), which is composed of a set of vertices V and a set of edges E. The vertex and edge are associated with an attribute label, which is selected from a set of labels. The size of the graph is defined as the number of edges in the graph. If a graph is called an i-subgraph, then it contains i edges. The graph database $DB = \{G_1, G_2 \ldots G_n\}$ is a collection of graphs, where each graph represents a single record. The definition p is a subgraph, and the support of p is defined as support(p), which means the number of p subgraphs contained in DB. Given the minimum support threshold θ, if and only if $\theta \leqslant support(p)$, p belongs to the frequent subgraph. Determining whether a graph is a subgraph (or hypergraph) of another graph is called the subgraph isomorphism problem, which is known to be NP-complete [30].

Frequent subgraph mining is to find all frequent subgraphs whose support is greater than or equal to the minimum support threshold. This article mainly studies undirected graphs, but the current solution can be extended to the case of directed graphs.

The gSpan [5] is a frequent subgraph mining algorithm based on the depth search of graph, and it can make the expression of graph unique and easy to operate. The frequent DFS coding tree is constructed by expressions, and the whole is divided into multiple sub-problems to realize the recognition of frequent subgraphs, which has great parallelism and scalability and not only simplifies the mining task but also improves its efficiency. The algorithm finds frequent substructures without candidate generation. In detail, gSpan uses a five-tuple (i, j, L_i, $L_{(i, j)}$, L_j) to encode the edges in the graph, where L_i and L_j are the attribute labels of points V_i and V_j, $L_{(i, j)}$ is the label of the edge between the points V_i and V_j, and the graph can be described with a group of such codes, which is called the DFS code. And an example is given to illustrate the specific implementation process of the algorithm in this section. In Fig.1 a [31], the edge of (V_0, V_1) is represented as (0, 1, X, a, X), and the identification subscript of DFS coding corresponding to b, c, d in Table 1[31], which are isomorphic. The earlier the graph is found, the smaller the DFS code subscript is. Therefore, the minimum DFS code is D_0, according to $D_0<D_1<D_2$. At the same time, an important theorem is shown in Theorem 3.

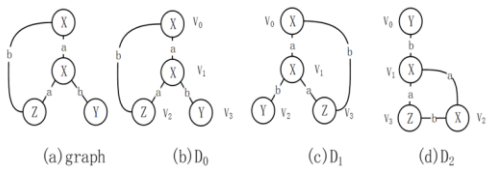


Fig. 1. Three DFS trees for a graph.

TABLE I. DFS CODES

DFScodes	D_0	D_1	D_2
edge e_0	(0,1,X,a,X)	(0,1,X,a,X)	(0,1,Y,b,X)
edge e_1	(1,2,X,a,Z)	(1,2,X,b,Y)	(1,2,X,a,X)
edge e_2	(2,0,Z,b,X)	(1,3,X,a,Z)	(2,3,X,b,Z)
edge e_3	(1,3,X,b,Y)	(3,0,Z,b,X)	(3,1,Z,a,X)

Theorem 3: (Subgraph isomorphism [5]). If two graphs are isomorphic, their minimum DFS codes are the same.

The core idea of the gSpan algorithm is as follows:(1) Calculate the support of all edges in the input graph set, delete all infrequent edges from the input graph set, and use frequent unilateral edges as the initial subgraph. (2) Extend frequent subgraphs. (3) Calculate the support degree of the candidate subgraph, save it if it is greater than the minimum support degree, otherwise delete it. (4) Delete the nonstandard code. (5)Reduce the graph set.

A Pattern Growth-based frequent subgraph mining algorithm that satisfies ε-differential privacy can improve the efficiency and scalability of the algorithm effectively. Therefore, we put forward the DP-gSpan.

IV. DP-GSPAN

The input content of the FSM algorithm is a graph database, such as a set of personally contributed records (e.g., a trajectory graph), and the final output generated frequent subgraphs and their supports. However, directly releasing the frequent subgraphs will pose considerable concerns on the personal privacy of the users participating in the data. We will construct a DFS coding tree under differential privacy as search space for DP-gSpan in this paper, and all frequent subgraphs can be obtained from the tree. By applying differential privacy in FSM, the adversary cannot infer the presence or absence of any personal records in the result. Therefore, the privacy of individuals can be protected through the results. The specific process and implementation details of DP-gSpan are described in this section.

A. Algorithm Description

The paper proposes a depth-first two-phase frequent subgraph mining algorithm under differential privacy, called DP-gSpan. DP-gSpan includes the frequent subgraph identification phase and the noisy support computation phase. In the first stage, a truncation strategy is needed to control the size of the search space. In the second stage, a newly designed privacy budget allocation strategy is adopted to improve the accuracy of noise support. The global algorithm description is shown in Fig.2.

In the first phase of DP-gSpan, an approximate frequent subgraph coding tree is constructed according to the core idea of gSpan(Algorithm 3). The tree is composed of many subtrees. An example of DP-gSpan used for frequent subgraph discovery is shown in Fig.3. Random noise is added in support, and the candidate frequent subgraph is obtained after the noise support is compared with the threshold. The algorithm starts mining from candidate frequent 1-subgraphs containing one edge until no candidate frequent i-subgraphs containing this edge are generated, and then we delete edges from the data set to reduce the data set. Since narrowing the candidate set helps to improve the accuracy of mining frequent subgraphs and the amount of noise increased by the differential privacy in the second stage, which is proportional to the number of candidate sets. Therefore, it is necessary to adopt a truncation strategy in the tree to reduce the number of candidate frequent subgraphs. According to empirical research, heuristic methods are used to retain 85% of the data to control the depth of the tree. The algorithm obtains the maximum depth of the tree by truncating branches with larger depths directly. In this approach, we only output the selected i-subgraphs and their real supports and do not output their noisy supports.

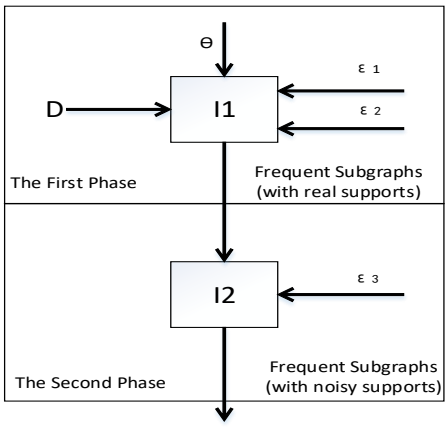


Fig. 2. An overview of our DP-gSpan algorithm.

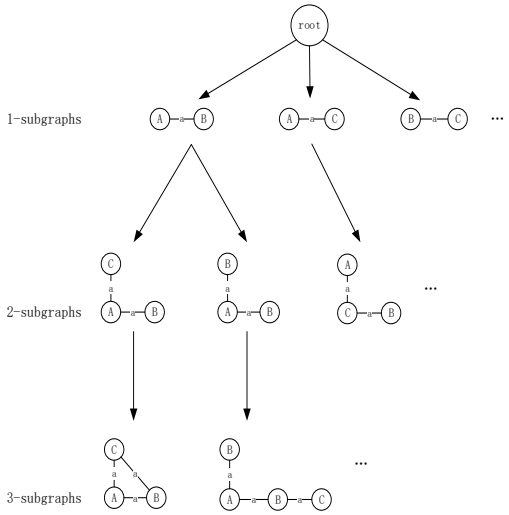


Fig. 3. The subgraph generation process.

After privately identifying frequent subgraphs, the tree is split into columns in the second phase. The deeper the branch contains, the greater the privacy budget required. Therefore, it is necessary to design a new privacy budget allocation strategy.

Algorithm 1 describes the overall process of the DP-gSpan. In line 1, the gSpan mining method is used to generate the approximate frequent subgraphs of the graph database D. In order to generate candidate frequent subgraphs, this paper proposes to add any amount of noise after calculating the true support and determine whether it is a candidate frequent

Algorithm 1 DP-gSpan Algorithm

Input: Graph database D, Threshold θ, Privacy budgets ε_1、ε_2 and ε_3（$\varepsilon_1+\varepsilon_2+\varepsilon_3=\varepsilon$）;
Output: A set F of frequent subgraphs together with their noisy supports;

1: Candidate Frequent Subgraph DFS coding tree D'←gSpan(D, θ, ε_1);
2: maxdepth L ← TruncateDatabase(D');
3: F←frequent_subgraph_identification(D', L, ε_2);
4: privacy_budget=PrivacyAllocation(F, ε_3);
5: noisy_support_computation(F, privacy_budget);
6: **return** F;

subgraph by the noise support. Finally, a candidate frequent DFS coding tree D' is generated. We use the heuristic-based

truncation function in the second line to determine L the maximum depth of the tree. In line 3, combines differential privacy mechanism to select frequent subgraphs from the truncated DFS coding tree and output frequent subgraphs and their real supports. In line 4 to line 5, adds noise to its real support according to the weight of the filtered frequent subgraphs combined with differential privacy according to the designed privacy budget allocation strategy, and finally returns all the frequent subgraphs and their corresponding noise supports.

B. Privacy Budget Allocation Strategy

DP-gSpan is a frequent subgraph mining algorithm under differential privacy that uses the DFS coding tree of frequent candidate subgraphs as the search space. The traditional privacy budget allocation method is to allocate the privacy budget to each column of the tree evenly, but this will lead to an imbalance in the signal-to-noise ratio of branches with different depths. Therefore, it is necessary to allocate different privacy budgets for columns of different depths of the tree according to the characteristics of the tree.

Except for the root node, the total number of columns with different depths in the frequent subgraph DFS coding tree is defined as m. According to the column depth, which is sorted from small to large, weights $w_i = \frac{1}{m-(i-1)}$ are assigned to columns of different depths in sequence, where i is the sorted column label. Start with the first column, and then the current number of columns after the first column. The total weight W is:

$$W = w_1 + w_2 + w_3 + \ldots + w_{i-1} + w_i = \frac{1}{m} + \frac{1}{m-1} + \frac{1}{m-2} + \ldots + \frac{1}{m-i+2} + \frac{1}{m-i+1} \quad (4)$$

Using a new privacy budget allocation strategy can achieve the effect of diminishing privacy budget in the frequent subgraph DFS coding tree according to the depth of each branch from large to small, which is more in line with the characteristics of the frequent subgraph DFS coding tree. Each column will be assigned $\varepsilon_i = \frac{\varepsilon}{W} \times w_i$, which is a specific privacy budget.

C. Truncation Strategy

In DP-gSpan, another important factor that affects the amount of noise in support is the sensitivity of the query function. The sensitivity in frequent subgraph mining is the same as the number of candidate subgraphs. For example, the query function needs to query the support of an i-subgraph, and the corresponding sensitivity is the number of all candidate i-subgraphs. Therefore, reducing the number of candidate subgraphs can reduce the sensitivity of the query function effectively. It is found that the noise support of the DFS coding tree decreases gradually with the increase of depth until all of the frequent subgraphs are mined with the minimum support that satisfies the threshold requirement. After the candidate subgraphs are generated, heuristics are used to truncate them. Details will be introduced later.

In Algorithm 2, $z_1 \ldots z_n$ is the statistics of the depth from small to large in the candidate frequent subgraph DFS coding tree D' from line 1. For example, z_1 represents the total number of branches with the smallest depth. Lines 2 to 7 use heuristic methods to obtain the maximum depth L that satisfies the requirements and the truncated branches are stored in D'' until the depth of all branches is within a limited range. The estimation function from lines 8 to 11 is the statistics of the

Algorithm 2 TruncateDatabase

Input: Frequent Subgraph DFS coding tree D';
Output: DFS coding tree of truncated frequent subgraphs D'';

```
1:  <z1,z2……,zn>=EstimateNum(D');
2:  L←the smallest integer such that Σ_{i=1}^{l} z_i ≥ 0.85;
3:  D''=NULL;
4:  for each path d∈D' do
5:      add d'=Truncate(L, d) to D'';
6:  end for
7:  return D'';
8:  function EstimateNum(D')
9:      Let z=<z1,z2……, zn> where zi is the # of paths
          with depth i in D';
10:     return z'/m where m is the # of total number paths
          in D;
11: end function
12: function Truncate(L, d)
13:     d'=min{L, |d|};
14:     d'={pick d' graphs from d};
15:     return d';
16: end function
```

number of branches contained in different depths and calculates the percentage of the number of branches containing the depth. Lines 12 to 16 are the pruning truncation function. If the depth of the branch is greater than the maximum depth L, the part of the branch greater than the depth L is directly truncated, otherwise, the original depth is retained |d|. Finally, all the processed branches d' are returned.

Algorithm 3 gSpan Algorithm

Input: Graph database D, Threshold θ, Privacy budgets ε_0;
Output: A set F of frequent subgraphs;

```
sort the labels in D by their noise frequency;
remove infrequent vertices and edges;
relabel the remaining vertices and edges;
S'←all frequent 1-edge graphs in D;
sort S' in DFS lexicographic order;
S←S';
for each edge e∈S' do
  initialize s with e, set s.D by graphs which contains e;
  Subgraph_Mining(D, S, s);
  D←D-e;
  if |D|<minSup then
    break;
  end if
end for
function Subgraph_Mining(D, S, s)
  if s≠min(s) then
    return;
  else
    S←S∪{s};
    enumerate s in each graph in D and count its
      children;
    for each c, c is s' child do
      if noisy support(c)≥minSup then
        s←c;
        Subgraph_Mining(D, S, s);
      end if
    end for
  end if
end function
```

D. Frequent Subgraph Mining

Algorithm 3 is a detailed description of the gSpan algorithm. In order to generate candidate frequent subgraph DFS coding trees, some changes are made to the original gSpan algorithm, and a certain amount of noise is added when digging out the frequent edges firstly and then if it is also the minimum DFS coding, expand gradually from the 1-subgraph until there is no frequent subgraph that contains the edge. This edge is deleted in the graph database to reduce the data set, and the subgraph obtained after screening is added as a node to the coding tree during the mining process, and its true support is recorded at the same time.

V. ANALYSIS

A. Privacy Analysis

Theorem 3: The DP-gSpan algorithm satisfies ε-differential privacy.

Proof: In DP-gSpan, it can be divided into two phases. In constructing the DFS coding tree, DP-gSpan adds Laplace noise to the support of the subgraph to generate a candidate set. Because adding or reducing a graph in the database has the greatest impact on the support of this subgraph is 1, which achieves ε_1-differential privacy. After the DFS coding tree is obtained, the frequent subgraphs are filtered by the differential privacy mechanism. Because adding or deleting a node in the tree, the maximum impact on the support of each candidate frequent subgraph is 1, so its corresponding sensitivity is proportional to M the number of candidate frequent subgraphs after pruning. Therefore, the noise of $Lap(\frac{M}{\varepsilon_2})$ is added to the support of each obtained candidate frequent subgraph, which achieves ε_2-differential privacy.

In the second phase of DP-gSpan, DP-gSpan uses a privacy budget allocation strategy to assign weights to each column of the frequent subgraph DFS coding tree with different depths, the privacy budget corresponding to each column except the root node is :

$$\varepsilon_{3i} = \frac{\varepsilon_3}{W} \cdot w_i = \frac{\varepsilon_3}{W} \cdot \frac{1}{m-(i-1)} \tag{5}$$

The total privacy budget consumed at this phase is:

$$\varepsilon = \frac{\varepsilon_3}{W} \cdot \frac{1}{m} + \frac{\varepsilon_3}{W} \cdot \frac{1}{m-1} + \frac{\varepsilon_3}{W} \cdot \frac{1}{m-2} \ldots + \frac{\varepsilon_3}{W} \cdot \frac{1}{m-(i-1)} \tag{6}$$
$$= \frac{\varepsilon_3}{W} \cdot \left(\frac{1}{m} + \frac{1}{m-1} + \frac{1}{m-2} + \cdots + \frac{1}{m-(i-1)}\right) = \frac{\varepsilon_3}{W} \cdot W = \varepsilon_3$$

When DP-gSpan mines each column, the sensitivity of the support for each frequent subgraph in the column is the number of nodes in the column and adding or deleting a node will affect its support at most 1, so the Laplace noise $Lap\left(\frac{\Delta f}{\varepsilon_{3i}}\right)$ added to each node satisfies ε_{3i}-differential privacy. Combining formula (6) and the principle of sequence combination, it can be seen that the second phase of DP-gSpan achieves ε_3-differential privacy.

According to the principle of sequence combination, DP-gSpan satisfies $(\varepsilon_1 + \varepsilon_2 + \varepsilon_3)$ -differential privacy. Through the above analysis, it can be proved that under a given privacy budget, DP-gSpan can achieve sufficient privacy guarantee for frequent subgraph mining without damaging the frequent subgraph itself.

TABLE II. DATASETS

Dataset	#Graphs	Avg.size	Max.size	#Vertex Labels
Cancer	32557	28.3	236	67
HIV	42689	27.5	247	65

VI. EXPERIMENTS

This paper compares DP-gSpan with the two algorithms of frequent subgraph mining under differential privacy proposed in recent years. The first algorithm is the DFS algorithm proposed by Cheng [8] and others, which also has two phases. The second algorithm is that Shen et al. [7] proposed an algorithm based on Markov Chain Monte Carlo (MCMC) sampling to realize frequent graph pattern mining based on differential privacy. These two algorithms are called DFG and DFPM respectively. Because the protection algorithm based on differential privacy is random, each experiment is performed 10 times, and then the average of all results is taken as the final result. In this experiment, the fixed privacy budget of experiments is 0.2. we set ε_1=0.01ε, ε_2=0.5ε, and ε_3=0.49ε. We implement all these algorithms in Java with Intel Core i5 CPU 1.38 GHz and 16 GB RAM, running Windows 10.

A. Datasets

In order to evaluate the performance of DP-gSpan in many aspects, this paper conducts experiments on the following two commonly used real data sets: dataset Cancer includes structures of human tumor cell lines, dataset HIV contains molecules tested against HIV virus. The specific details of the data set are shown in Table 2.

B. Utility Measures

In the experiment, this paper uses F-Score [8] and RE two general utility evaluation indicators to evaluate the algorithm proposed in this paper. When the F score is larger, it means that the published frequent subgraph is closer to the original true frequent subgraph. The smaller the RE, the smaller the error generated by the noise support, and the better the algorithm effect.

F-score is used to measure the accuracy of mining results, as shown in equation (7):

$$F - Score = 2 * \frac{precision * recall}{precision + recall} \tag{7}$$

Where, $precision = \frac{|F \cap F\prime|}{F\prime}$, and $recall = \frac{|F \cap F'|}{F}$。 F is the accurate result of frequent subgraph mining, F' is the result of frequent subgraph mining under differential privacy.

RE is used to measure the error concerning the true supports, as shown in equation (8):

$$RE = median_{x \in F'} \frac{sup'_x - sup_x}{sup_x} \tag{8}$$

sup'_x is the support after adding noise to the subgraph, and sup_x is the original support of x.

C. The Influence of Threshold

Fig.4 and Fig.5 show the performance of DP-gSpan, DFG, and DFPM on Cancer and HIV under different thresholds. In DFG, the same privacy budget allocation scheme as Cheng [8] is adopted, ε_1=0.1ε, ε_2=0.5ε, and ε_3=0.4ε. The F-Score of DP-gSpan on the data set is better than DFG and DFPM. This is because DP-gSpan uses a heuristic truncation strategy in the candidate frequent subgraphs. Cutting out the smaller support of subgraphs in the DFS coding tree, which are likely to be infrequent subgraphs can greatly reduce the candidate set and less noise can be added to protect user privacy in the second phase. It can be found that DFG is superior to DP-gSpan and DFPM in RE. This because DFG uses a heuristic method of

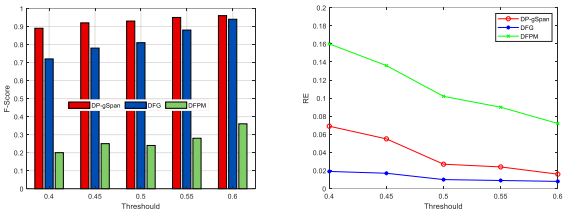
constructing error perception paths during adding noise to the support, which minimizes the variance of the noise added by each node through the method of noise accumulation. However, the RE of DP-gSpan is better than DFPM on both data sets, so it can be proved that the DP-gSpan proposed in this paper effectively improves the utility of the results.

D. The Influence of Privacy Budget

The performance of the three algorithms for mining top-50 frequent subgraphs under varying privacy budgets on datasets Cancer and HIV is shown in Fig.6 and Fig.7, and the results of privacy budgets between 0.1 and 0.5. It can be seen that with the increase of privacy budget, the F-Score and RE of all algorithms have a trend of getting better gradually. This is because increasing the privacy budget can reduce the amount of noise introduced, thereby improving the quality of the frequent subgraphs gradually in the experimental results. We can easily see that in terms of F-Score, DP-gSpan has a better performance than DFG and DFPM. In terms of RE, DP-gSpan is more accurate than DFPM significantly. This is because DP-gSpan uses a new privacy budget allocation strategy to allocate a more appropriate privacy budget for each column, making the results more usable.

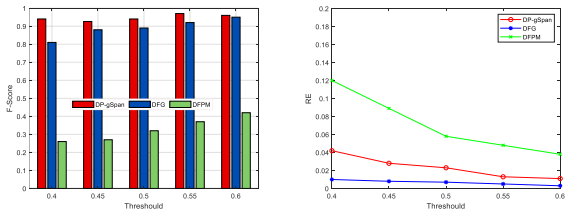
E. The Influence of K Value

Fig.8 and Fig.9 show the performance of DP-gSpan, DFG, and DFPM in the Top-k frequent subgraph mining experiment with different k values. It can be seen from the figures that as the value of k increases, the F-Score of all algorithms shows a downward trend, but DP-gSpan still performs better in all data sets compared to DFG and DFPM.



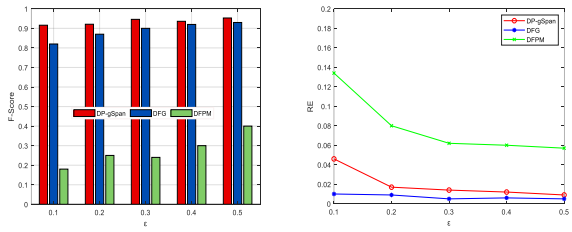
(a) Cancer : F-Score (b) Cancer : RE

Fig. 4. The impact of threshold.



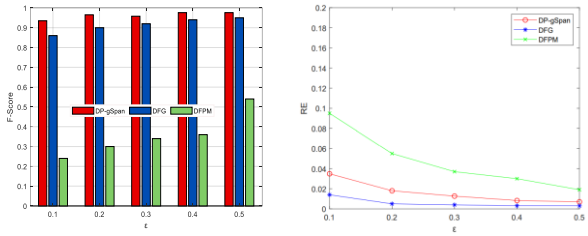
(a) HIV : F-Score (b) HIV : RE

Fig. 5. The impact of threshold.



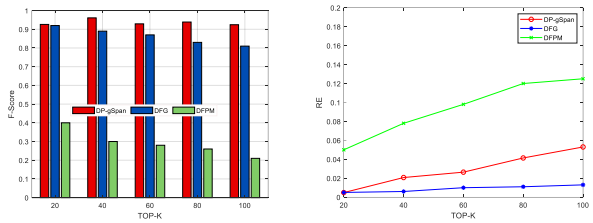
(a) Cancer : F-Score (b) Cancer : RE

Fig. 6. The impact of privacy budgets.



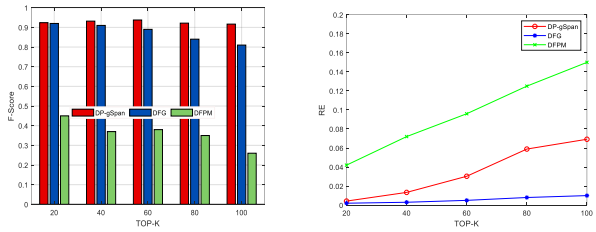
(a) HIV : F-Score (b) HIV : RE

Fig. 7. The impact of privacy budgets.



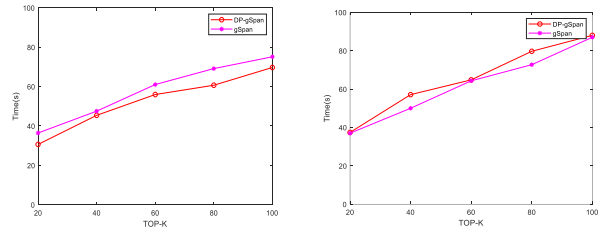
(a) Cancer : F-Score (b) Cancer : RE

Fig. 8. The impact of k value.



(a) HIV : F-Score (b) HIV : RE

Fig. 9. The impact of k value.



(a) Cancer (b) HIV

Fig. 10. Running time.

F. Running Time

Fig.10 shows the performance of DP-gSpan and gSpan in the Top-k frequent subgraph mining experiment with different k values. The gSpan algorithm is a frequent subgraph mining algorithm without privacy protection. It can be seen from the figures that as the value of k increases, the running time gradually increases. The running time of the gSpan algorithm and the DP-gSpan algorithm is similar on Cancer and HIV data sets. This is due to the computing the support of graphs is the most time-consuming part, which has exponential time complexity. For DP-gSpan, it needs an extra phase (i.e., a heuristic truncation process of the frequent subgraph identification phase) to determine the depth of the DFS coding

tree. Therefore, the number of reserved frequent subgraphs is uncertain and the computation cost during the second phase of the noisy support of identified frequent subgraphs will fluctuate with it, but it will do not introduce too much difference.

VII. CONCLUSION

We propose a new FSM algorithm under differential privacy called DP-gSpan, which further improves the data utility and mining efficiency while ensuring data security. This algorithm consists of a frequent subgraph identification phase and a noisy support computation phase. Different from the previous Apriori-based algorithms, DP-gSpan does not produce many redundant candidate sets, which will save a lot of running time compared to the previous ones. Moreover, a heuristic truncation strategy and a new privacy budget allocation strategy, which improve the trade-off between data utility and privacy. Privacy analysis also proves that DP-gSpan satisfies ε-differential privacy effectively. According to the comparative experimental results on multiple real data sets, DP-gSpan has better performance than the previous algorithms.

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